

Two-Step Equations in Context*Additional Practice 7A.01.04*

Name _____

Date _____

Period _____

Lisa solved the equation $-2x - 3 = 5$. Her work is shown. Use it for question 1.

$$-2x - 3 + 3 = 5 + 3$$

$$-2x = 8$$

$$-2x \div (-2) = 8 \div (-2)$$

$$x = 4$$

1. Use substitution to decide whether Lisa's solution is correct. If it is not, describe her error and solve the equation correctly.

Zoey solved the equation $3m + 5 = 8$. Her work is shown. Use it for question 2.

$$3m + 5 - 5 = 8 - 5$$

$$3m = -3$$

$$3m \div 3 = -3 \div 3$$

$$m = -1$$

2. Use substitution to decide whether Zoey's solution is correct. If it is not, describe her error and solve the equation correctly.

For questions 3 through 5, solve the equation, then say what the solution means in the situation.

3. A taxi charges \$1.75 plus \$0.75 for each mile traveled. Your trip cost \$14.50. The equation $1.75 + 0.75x = 14.50$ represents the situation.

4. A zipline course charges a flat rate of \$75 plus \$5.50 for each line you cross. One person's total was \$124.50. The equation $75 + 5.50z = 124.50$ represents the situation.

5. A runner warms up for 15 minutes and then runs each mile in 6 minutes. The total time is 45 minutes. The equation $15 + 6m = 45$ represents the situation.

For questions 6 through 10, solve the equation and check your solution by substitution.

6. $125 = 15 + 5x$

7. $9.2 = 6.1 + \frac{h}{6}$

8. $17.5c - 15 = 98.75$

9. $\frac{t - 50}{4} = 400$

10. $16.42 + 2.51r = 26.46$